

REMOTE WORK AND LIFE SATISFACTION IN THE UK: GEOGRAPHIC AND WITHIN-GENDER HETEROGENEITY

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Abstract

This paper examines heterogeneous effects of the post-pandemic shift to remote work on wellbeing in the United Kingdom. Exploiting pre-pandemic occupational teleworkability as a source of plausibly exogenous variation and applying the doubly robust differences-in-differences estimator of Callaway and Sant’Anna (2021), we extend the aggregate framework of Cowan and Spearing (2026) to examine effect heterogeneity across UK regions and within gender subgroups defined by partnership and parental status. Using UK Household Longitudinal Study data spanning 2014 to 2023, we find suggestive evidence that aggregate null effects on mental health conceal meaningful variation: single mothers exhibit the largest adverse point estimates for GHQ-12 caseness, and workers outside London show worse mental health trajectories than those in metropolitan labour markets. Wide confidence intervals counsel against strong conclusions, but the findings carry implications for the geographic targeting of mental health support and the design of flexible working arrangements sensitive to household composition.

Keywords: Remote work, wellbeing, mental health, heterogeneous treatment effects, regional inequality, gender, United Kingdom

1 Introduction

The labour market consequences of the COVID-19 pandemic are well documented, but one of its most lasting impacts remains underexplored: the structural shift towards remote work and its distributional effects on wellbeing. What was once a marginal employment arrangement now characterises the working lives of a substantial share of the British workforce. Cowan and Spearing (2026) document that workers in teleworkable occupations experienced large and sustained increases in the probability of working remotely following 2020, with over half of UK workers now sometimes working from home, a fourfold increase from pre-pandemic levels. Whether this shift has been benign in its consequences for population wellbeing remains contested.

A tension sits at the heart of this literature. Workers report valuing the flexibility and autonomy that remote arrangements afford, with some willing to accept modest wage reductions to retain work-from-home options (Mas and Pallais, 2017). Yet the reduction in face-to-face workplace contact may undermine social connection in ways that harm mental health over the longer run. These competing forces need not cancel out uniformly. Their net effect likely depends on who is working remotely, where they live, and what social infrastructure surrounds them outside the office. The existing literature, whilst advancing considerably since 2020, has largely treated these dimensions as secondary to aggregate average effects.

This paper directly addresses that gap. We extend the empirical framework of Cowan and Spearing (2026) to examine heterogeneous impacts of remote work on life satisfaction, loneliness, and mental health across geographic regions and within gender groups in the United Kingdom. Our contribution is to examine the geographic and intra-gender dimensions of effect heterogeneity. Cowan and Spearing find that aggregate adverse effects were transient and small. Our question is whether those aggregate null results conceal meaningfully different experiences across regions and population subgroups, with important implications for spatially targeted policy and the design of flexible working arrangements. We find suggestive evidence that they do, particularly among single mothers and workers outside London, though wide confidence intervals counsel against strong conclusions.

Our identification strategy exploits pre-pandemic occupational teleworkability as a source of plausibly exogenous variation in remote work exposure, using the doubly robust differences-in-differences estimator of Callaway and Sant’Anna (2021). Within this framework, we extend the analysis to examine effect heterogeneity by UK region, and within gender categories, distinguishing between partnered and unpartnered women with and without dependent children, and equivalent distinctions within the male labour force.

2 Literature Review

2.1 Remote Work and Wellbeing

The pandemic created a large-scale natural experiment in remote work adoption, and a substantial literature has emerged to exploit it. Early cross-sectional studies documented mixed effects on job satisfaction and mental health (Galanti et al., 2021; Song et al., 2020), but these are largely unable to address selection: workers who remote work may differ from office workers in ways correlated

with wellbeing. Panel studies using individual fixed effects improve on this, but remain vulnerable to reverse causality if mental health conditions themselves influence remote work decisions (Bilgrami et al., 2023; Esposito and Terlizzese, 2024).

The most credible causal evidence uses plausibly exogenous shocks to remote work adoption. Goux et al. (2025) exploit the staggered introduction of firm-level remote work agreements in France, finding deterioration in health measured across chronic conditions and self-reported poor health. Bertoni et al. (2021) identify negative effects on loneliness and mental health during the acute pandemic period for older European workers, using pre-pandemic occupational teleworkability interacted with time as their identifying variation. Costi et al. (2024) study the reverse shock (a mandated return-to-office policy for Italian public sector workers in 2021), finding null effects on loneliness and mental health.

Cowan and Spearing (2026) represents the most directly relevant antecedent to the present study and is the paper upon which we build. Using UK Household Longitudinal Study (UKHLS) data spanning 2011 to 2023, they compare workers in teleworkable occupations to those in non-teleworkable roles using the doubly robust estimator of Callaway and Sant’Anna (2021). Their headline finding is that whilst teleworkable workers experienced a 15 percentage point increase in the probability of working mostly from home following 2020, adverse effects on self-reported loneliness and mental health were transient. Event studies reveal elevated loneliness in 2021–2022 that subsides by 2023, and no statistically significant post-pandemic effects on GHQ12 or SF12 mental health indices. Effects on job satisfaction are similarly negligible, though there is suggestive evidence of improved work-related autonomy for men under the synthetic differences-in-differences estimator. Heterogeneity analyses by marital and parental status find that married individuals and parents exhibit somewhat larger loneliness effects, though these are inconsistent across specifications. Crucially, Cowan and Spearing do not examine geographic heterogeneity, and their within-gender analysis remains coarse.

A broader pattern emerges from the literature: studies focusing on the acute pandemic period tend to find negative associations between remote work and mental health, whilst those extending into the post-pandemic period report attenuation (Senik and Postel-Vinay, 2024; Gueguen and Lhuillier, 2023; Kornadt and Schmidt, 2025). This temporal dynamic is consistent with an adjustment process in which workers either adapt to changed conditions or self-select into arrangements—hybrid or otherwise—that suit them better. It does not preclude the possibility that adjustment has been uneven across geographic and demographic groups.

2.2 Gender, Family Structure, and Differential Effects

The gender dimension of remote work’s wellbeing consequences has attracted attention, though findings are heterogeneous. Arntz et al. (2022) find that the labour supply response to remote work opportunities varies substantially by household composition, with women bearing disproportionate care responsibilities that may compound the pressures of home-based work. Bilgrami et al. (2023) document gender differences in how isolation is experienced under remote working. Flexible working arrangements more broadly appear to operate differently for men and women depending on family circumstance (Agnoletto and Bonomi, 2024).

What the literature lacks is within-gender disaggregation. Treating “women” as a single analytical

category conflates meaningfully different circumstances: a partnered mother with school-age children navigates remote work differently from a single woman without dependants, both in terms of the domestic pressures remote work generates and the social functions that workplace contact might otherwise serve. Cowan and Spearing (2026) note heterogeneity by parental status – significant loneliness effects among fathers, less clear patterns among mothers—but their analysis does not pursue the interaction of partnership status and parental status within gender. This paper attends directly to this gap.

2.3 Geographic Dimensions

Remote work’s wellbeing consequences are likely geographically conditioned, yet this dimension has received almost no direct empirical attention. Workers in dense urban labour markets can exploit flexible arrangements whilst retaining access to thick professional and social networks; workers in more peripheral or economically disadvantaged regions may find that remote work amplifies isolation precisely because those networks are thinner. The UK context makes this especially salient: regional inequalities in productivity, amenities, and social infrastructure are pronounced, and the government’s levelling-up agenda makes the distributional geography of policy-relevant outcomes a live concern. If the aggregate null results of Cowan and Spearing (2026) mask regionally concentrated harms, that finding carries direct implications for how flexible working policy should be designed and targeted.

2.4 Methodological Background

Identifying causal effects of remote work on wellbeing poses a fundamental challenge: workers who select into remote arrangements may differ from office workers in ways correlated with mental health trajectories. The approach adopted here follows Cowan and Spearing (2026) in exploiting pre-pandemic occupational teleworkability as a source of variation that is determined by task characteristics prior to 2020 and is therefore unlikely to be endogenously related to pandemic-period wellbeing shocks. This strategy avoids the reliance on instrumental variables that may violate exclusion restrictions. Commuting distance, for instance, could plausibly affect wellbeing through channels independent of remote work itself.

The doubly robust estimator of Callaway and Sant’Anna (2021) provides the econometric framework. Rather than assuming unconditional parallel trends between teleworkable and non-teleworkable occupation groups, that differ substantially across education, income, and household composition, this estimator conditions on a rich set of pre-determined covariates to render trends parallel conditional on observables. The estimator is consistent for the causal effect provided either the propensity score or the linear outcome model is correctly specified. We apply this framework to our geographic and within-gender subgroup analyses, inheriting its robustness properties whilst extending the scope of heterogeneity examined.

2.5 Contribution

Our contribution is threefold. We provide the first systematic evidence on geographic heterogeneity in the remote work–wellbeing relationship within the UK, directly relevant to debates about whether flexible working amplifies or attenuates regional inequality. We extend within-gender

heterogeneity analysis to account for the interaction of partnership status and parental status, moving beyond the coarse male/female comparison that dominates the literature. In doing so, we find suggestive evidence that the aggregate null effects documented by Cowan and Spearing (2026) conceal heterogeneous impacts across population subgroups: single mothers exhibit the largest positive point estimate for GHQ-12 caseness (0.376), and workers in the South and North show worse mental health trajectories than those in London or the Midlands, consistent with the hypothesis that metropolitan workers can more readily substitute workplace social contact through denser external networks. Wide confidence intervals across all subgroup estimates counsel caution, but the dispersion in point estimates is large enough to carry meaningful policy implications.

3 Data

3.1 Understanding Society

Following Cowan and Spearing (2026) our primary data source is the UK Household Longitudinal Study (UKHLS), also known as Understanding Society, a large-scale longitudinal household panel survey conducted annually by the Institute for Social and Economic Research at the University of Essex. We use ten consecutive waves spanning 2014 to 2023 (waves f through o), which straddle the COVID-19 pandemic and allow us to observe both the pre-pandemic baseline period and the sustained post-pandemic shift in remote work patterns. The survey collects information on a wide range of individual characteristics, labour market outcomes, and wellbeing measures.

Our analytic sample is restricted to working-age individuals aged 25 to 64 who are employed in either full- or part-time employment and have a valid three-digit ISCO88 occupation code in at least one wave. Following Cowan and Spearing (2026), we anchor the treatment and control group classification to each individual’s occupation as observed in 2019, the last pre-pandemic wave, and exclude the acute pandemic years 2020 and 2021 from the outcome regressions to avoid conflating the permanent structural shift in remote work with the transitory disruptions of lockdowns, school closures, and furlough schemes. After imposing these restrictions and dropping observations with missing values on the outcome variables or conditioning covariates, the analytic sample comprises approximately 97,000 person-year observations, drawn from 12,938 unique individuals. Descriptive statistics for some chosen variables including mental health indicators (GHQ-12 and SF-12) are shown below in Table 1.

Table 1: Descriptive Statistics by Within-Gender Group

Group	% Teleworkable	GHQ-12	SF-12 MCS	N
<i>Women</i>				
Partnered mother	52.5	1.83	47.3	25,473
Partnered, no children	51.9	1.78	48.4	31,917
Single mother	38.5	2.44	44.8	5,209
Single, no children	50.0	2.23	46.2	17,374
<i>Men</i>				
Partnered father	51.0	1.39	49.0	28,719
Partnered, no children	47.4	1.23	50.4	29,727
Single father	45.8	2.11	47.3	452
Single, no children	45.2	1.60	47.9	14,034
<i>Full sample</i>				
All workers	51.8	1.67	48.3	153,274

Notes: Sample restricted to working-age (25–64) employed individuals across UKHLS waves f–o (2014–2023). GHQ-12 caseness ranges from 0 (no adverse symptoms) to 12 (maximum symptoms); higher values indicate worse mental health. SF-12 Mental Component Score ranges from 0 to 100; higher values indicate better mental health. % Teleworkable reports the share of each group working in teleworkable occupations as classified by Dingel and Neiman (2020) via the Hardy (2016) crosswalk. Partnership status is defined as married, civil partnership, or cohabiting. Single fathers ($n = 452$) should be interpreted with caution given the small sample size.

3.2 Occupational Teleworkability and Treatment Assignment

The central identification challenge is that workers who choose to work from home may differ from those who do not in ways correlated with their wellbeing. To circumvent this selection problem, we exploit variation in occupational teleworkability: the degree to which a job’s tasks can feasibly be performed remotely. Workers in highly teleworkable occupations experienced a far larger post-pandemic increase in remote work than those in non-teleworkable occupations, for reasons determined by task structure rather than by individual preference or circumstance.

We construct the teleworkability indicator using the occupation-level index of Dingel and Neiman (2020), which classifies occupations as teleworkable or non-teleworkable based on O*NET task descriptors. Following Cowan and Spearing (2026), we link respondents’ 2019 ISCO88 occupation codes to this index via the Hardy (2016) crosswalk.¹ Workers whose 2019 occupation falls into a teleworkable group constitute the treatment group with those in non-teleworkable groups constituting the control group. Of the 102,919 person-year observations with a valid 2019 occupation, 53,350 (51.8%) are classified as teleworkable (Table 2).

¹Because UKHLS under a standard licence provides occupation at the three-digit ISCO88 level rather than the four-digit level available under a special licence, we aggregate the binary teleworkability classification to the three-digit level by majority rule. This introduces measurement error relative to Cowan and Spearing (2026) and will attenuate our first-stage estimates towards zero and therefore our main results are therefore conservative.

Table 2: Characteristics of Workers in Teleworkable and Non-Teleworkable Occupations

	Non-Teleworkable	Teleworkable
<i>Wellbeing outcomes</i>		
Work mostly from home	0.034	0.181
GHQ-12 caseness	1.60	1.73
SF-12 MCS	48.5	48.1
Job satisfaction	5.38	5.38
Is lonely (%)	37.0	34.6
<i>Demographics</i>		
Age	45.8	44.8
Female (%)	53.0	53.8
Married or partnered (%)	75.5	78.8
Has children (%)	38.5	41.4
Degree or above (%)	39.7	68.2
Owns home (%)	73.6	84.6
White (%)	84.0	86.4
Mean monthly income (£)	1,827	2,541
Observations	49,569	53,350

Notes: The table reports mean values of key variables for workers in teleworkable and non-teleworkable occupations, pooled across UKHLS waves f-o (2014–2023). The sample is restricted to working-age (25–64) employed individuals observed in 2019. Teleworkability is classified using the Dingel and Neiman (2020) index linked to ISCO88 3-digit occupation codes via the Hardy (2016) crosswalk, aggregated by majority rule. GHQ-12 caseness ranges from 0 to 12 (higher = worse mental health). SF-12 Mental Component Score ranges from 0 to 100 (higher = better mental health). Job satisfaction is scored 1–7. Loneliness is a binary indicator equal to 1 if the respondent reports feeling lonely sometimes or often. Monthly income is expressed in nominal GBP.

3.3 Outcome Variables

We examine four wellbeing outcomes, all drawn from the UKHLS core questionnaire:

- **GHQ-12 Caseness:** The GHQ-12 is a series of 12 questions asking individuals about the severity of mental health symptoms, with four ordered responses normalised so that higher numbers reflect worse mental health (Goldberg, 1978). We construct the caseness score as the sum of symptoms on which a respondent scores 3 or 4, ranging from 0 (no symptoms) to 12 (maximum symptoms). In our sample, the mean caseness score is 1.60 for non-teleworkable workers and 1.73 for teleworkable workers.
- **SF-12 Mental Component Summary Score (MCS):** The SF-12 is a health status indicator derived from the 36-item Short Form Health Survey. We use the MCS Score, which summarises self-reported mental health across dimensions including vitality, social functioning, and emotional role limitation. The MCS is scaled 0 to 100, with higher values indicating better mental health and is available in a subset of UKHLS waves.
- **Job Satisfaction:** Respondents are asked to rate their overall satisfaction with their current job on a seven-point Likert scale from 1 (completely dissatisfied) to 7 (completely satisfied). Mean job satisfaction is 5.38 in both the teleworkable and non-teleworkable groups prior to the pandemic.
- **Loneliness:** This is an indicator variable equal to one if the respondent reports feeling lonely “sometimes” or “often” when asked about their frequency of loneliness. This item is available

from wave 9 (2017–18) onwards, meaning the pre-pandemic loneliness series is shorter than for the other outcomes. In our sample, 37.0% of non-teleworkable workers and 34.6% of teleworkable workers report feeling lonely at baseline.

3.4 Covariates

To support the conditional parallel trends assumption (see Section 4.1), we condition on a vector of pre-determined individual characteristics, all measured in 2019. These are: highest educational qualification (coded as two indicator variables for degree-or-above and A-level-equivalent, with below-A-level as the reference); an indicator for being married or cohabiting; an indicator for having dependent children; and an indicator for home ownership.² These variables are chosen because they differ systematically between teleworkable and non-teleworkable workers, as documented in Table 2, and because they are determined prior to the treatment period and therefore cannot themselves be outcomes of the work-from-home revolution.

3.5 Heterogeneity Variables

For the geographic heterogeneity analysis, we use the Government Office Region variable, anchored to 2019, to assign individuals to one of five regional groupings: the North (North East, North West, and Yorkshire and the Humber), the Midlands (East and West Midlands), London, the South excluding London (East of England, South East, and South West), and the Devolved nations (Wales, Scotland, and Northern Ireland).

For the within-gender heterogeneity analysis, we construct eight mutually exclusive subgroups from the interaction of gender, partnership (married, civil partnership, or cohabiting versus single), and parental status, all as observed in 2019. Single fathers are excluded from the main within-gender analysis due to insufficient sample size ($n = 452$), leaving seven subgroups. Descriptive statistics for these groups are reported in Table 1.

4 Empirical Strategy

4.1 Identification

Our empirical strategy follows Cowan and Spearing (2026) and exploits the COVID-19 pandemic as an exogenous shock that dramatically and differentially increased remote work opportunities across occupations. The identifying variation arises because workers in teleworkable occupations, as classified prior to the pandemic on the basis of task characteristics, experienced a far larger and more sustained increase in working from home after 2020 than workers in non-teleworkable occupations. This differential shift is visible in the first-stage event study (Figure 1). Prior to 2020, the two groups exhibit statistically indistinguishable rates of home working, while from 2020 onwards, a large and persistent gap opens up.

The key identifying assumption is conditional parallel trends: absent the pandemic-driven expansion of remote work, the trajectory of wellbeing outcomes would have evolved in parallel for teleworkable and non-teleworkable workers, conditional on the set of pre-determined covariates described in

²Home ownership is merged from the UKHLS household file and linked to individuals via the household identifier.

Section 3. Because teleworkable and non-teleworkable occupations differ systematically in education, income, and household structure (Table 2), we condition on these characteristics to rule out the possibility that such level differences reflect underlying trend differences. Crucially, since we use 2019 values of all conditioning variables, none of these controls can themselves be affected by the treatment. We additionally assume no anticipation: workers did not alter their wellbeing outcomes in advance of the pandemic shock, which is plausible given the suddenness and severity of the public health emergency in early 2020.

4.2 Estimator

Our preferred estimator is the doubly-robust differences-in-differences estimator of ?, which accommodates heterogeneous treatment effects across units and time periods and does not require a balanced panel.³ It computes group-time average treatment effects on the treated, $ATT(g,t)$, separately for each cohort g and calendar year t , using only not-yet-treated and never-treated units as the comparison group. In our setting, there is a single treatment cohort ($g = 2020$) and a never-treated control group (workers in non-teleworkable occupations). Following Sant’Anna and Zhao (2020), the group-time ATT is estimated via:

$$\widehat{ATT}(g,t) = E \left[\left(\frac{D_i}{E[D_i]} - \frac{\hat{p}(X_i)(1 - D_i)}{(1 - \hat{p}(X_i)) E[D_i]} \right) (Y_{it} - Y_{i,pre} - \hat{m}_{0,t}(X_i)) \right] \quad (1)$$

where $D_i \in \{0,1\}$ indicates membership in the treatment group (teleworkable occupation in 2019); $\hat{p}(X_i)$ is the estimated propensity score from a logit regression of D_i on the pre-determined covariate vector X_i ; $Y_{i,pre}$ is the outcome in the last pre-treatment period (2019); and $\hat{m}_{0,t}(X_i)$ is the predicted outcome change from 2019 to year t for control-group individuals with characteristics X_i , obtained from OLS on the control group. The estimator is *doubly robust*, meaning that it is consistent for the true ATT if either the propensity score model $\hat{p}(\cdot)$ or the outcome regression $\hat{m}_{0,t}(\cdot)$ is correctly specified (Sant’Anna and Zhao, 2020), providing insurance against functional-form misspecification relative to standard TWFE.

We present dynamic event-study estimates for each year 2014–2023 relative to 2019, and a summary ATT aggregated over 2022–2023, excluding 2020–2021 to avoid contamination from the acute pandemic period. The validity of the identifying assumptions is assessed in Section 4.3.

4.3 Validation of Identifying Assumptions

4.3.1 First Stage

As a precondition for our analysis, we verify that the teleworkability classification predicts a meaningful and sustained increase in working from home after 2020. Estimating equation eq:dr with working mostly from home as the outcome and excluding 2020–2021 from the regression, we obtain a summary ATT of 0.139 (bootstrapped SE = 0.010), indicating that the work-from-home revolution increased the probability of working mostly from home by approximately 14 percentage points among workers in teleworkable occupations relative to their non-teleworkable counterparts.

³Under treatment effect heterogeneity, standard two-way fixed effects estimators produce weighted averages of group-time treatment effects with potentially negative weights, which can yield sign-reversed estimates of the aggregate ATT (Goodman-Bacon, 2021).

The event study in Figure 1 confirms that pre-2020 differences are flat and indistinguishable from zero, while the post-2020 gap is immediate and persistent.

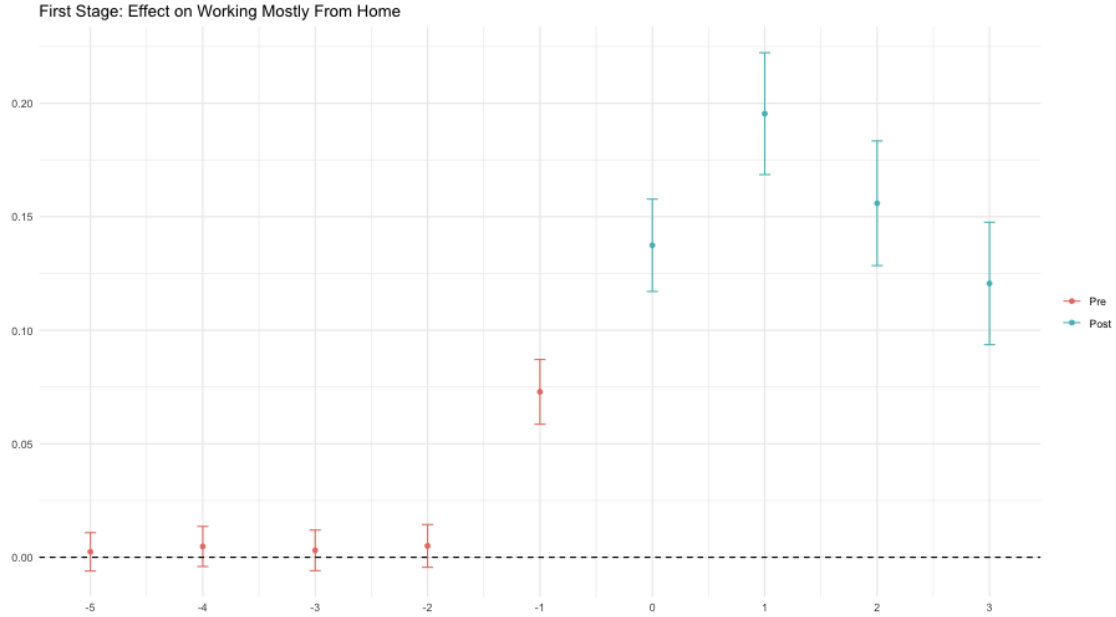


Figure 1: Callaway–Sant’Anna (2021) dynamic event study: effect of teleworkable occupation on probability of working mostly from home.

Estimates are relative to the 2019 baseline. Pre-treatment coefficients ($t < 2020$) are jointly indistinguishable from zero. Post-2020 gap is immediate and persistent. Years 2020–2021 are included for illustrative purposes only and are excluded from the summary ATT.

4.3.2 Pre-Treatment Balance

Table 3 reports balance tests for five pre-determined characteristics. All ATT estimates are economically negligible and statistically insignificant, with pre-trend p-values well above conventional significance thresholds, providing no evidence of differential pre-existing trends consistent with the conditional parallel trends assumption.

Table 3: Balance Test: Effect of Treatment on Pre-Determined Characteristics

Variable	ATT	SE	95% CI	Pre-trend p
Female	−0.009	0.004	[−0.017, −0.001]	0.350
Age	0.034	0.086	[−0.134, 0.202]	0.218
GHQ-12 Caseness	0.047	0.057	[−0.064, 0.158]	0.878
SF-12 MCS	−0.023	0.152	[−0.321, 0.275]	0.843
White	−0.010	0.003	[−0.016, −0.004]	0.393

Notes: Each row reports the aggregate ATT from a separate doubly-robust Callaway–Sant’Anna (2021) estimator in which the listed characteristic is the dependent variable. The treatment group comprises workers in teleworkable occupations in 2019 and the control group in non-teleworkable occupations. Conditioning covariates are marital status, having dependent children, home ownership, and educational qualification, all measured in 2019. The pre-trend p -value reports the result of a joint Wald test that pre-treatment coefficients are jointly zero. 95% confidence intervals are constructed as $ATT \pm 1.96 \times SE$. Standard errors are bootstrapped (1,000 replications). All estimates exclude 2020–2021.

4.3.3 Pre-Trend Test

Figure 2 plots the full event study for GHQ-12 caseness. A joint Wald test of the pre-treatment coefficients fails to reject parallel trends ($p = 0.878$),

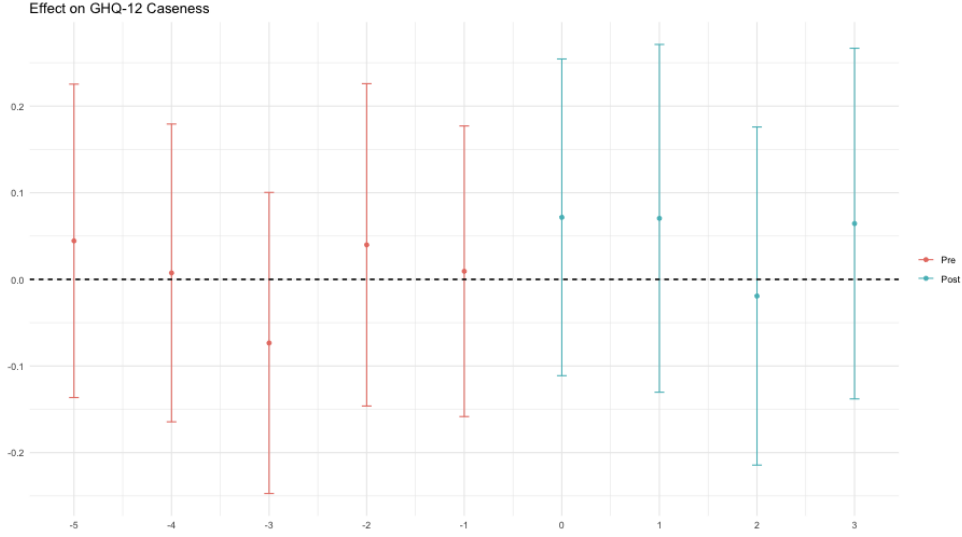


Figure 2: Callaway–Sant’Anna (2021) dynamic event study: effect of teleworkable occupation on GHQ-12 caseness.

Estimates are relative to the 2019 baseline. Pre-treatment coefficients ($t < 2020$) are jointly insignificant ($p = 0.878$), supporting the conditional parallel trends assumption. Shaded bands are 95% bootstrapped confidence intervals. Years 2020–2021 are shown for illustrative purposes only and excluded from the summary ATT.

4.4 Heterogeneity Analysis

To extend beyond current research, we re-estimate the doubly-robust DiD separately for five regional subsamples and seven gender-by-partnership-by-parenthood subgroups, as defined in Section 3. Within each gender subgroup regression, the partnership indicator is dropped from the conditioning vector to avoid collinearity.

4.5 Threats to Identification

There are a few concerns to identification worth discussing. First, occupational sorting: anticipatory selection into teleworkable occupations prior to 2020 would manifest as diverging pre-trends, which we do not observe. Second, spillovers to the control group: if the work-from-home revolution altered the experience of non-teleworkable workers through co-worker absence or changed workplace culture, our estimates represent a lower bound on the true effect. Third, panel attrition: if attrition is differential between groups, sample composition may change over time. The balance tests in Section 4.3 provide reassurance on this front.

5 Results

5.1 Baseline Results

Table 4 presents our main estimates of the effect of the work-from-home revolution on mental health and wellbeing outcomes, excluding the acute pandemic years 2020–2021. Across all four outcomes

the point estimates are small and statistically insignificant, with the exception of loneliness which is marginally significant at the 10% level.

The estimated effect on GHQ-12 caseness is 0.023, against a pre-treatment mean of 1.67. The pre-trend p -value of 0.878 provides strong support for the parallel trends assumption, and the event study in Figure 2 confirms that post-treatment coefficients are uniformly small and centred on zero. For the SF-12 MCS and job satisfaction indicators, their estimates are 0.028 and 0.024 respectively, both negligible relative to their pre-treatment means.

The loneliness estimate of 0.016 (SE = 0.010) is marginally significant at the 10% level, implying a 1.6 percentage point increase in the probability of reporting loneliness. Disaggregating by sex, the GHQ-12 point estimate is negative for men (-0.013) and positive for women (0.051), though neither is statistically significant and the confidence intervals overlap substantially; we examine this further in Section 4.4.

Table 4: Baseline Results: Effect of the Work-From-Home Revolution on Wellbeing

	GHQ-12	SF-12 MCS	Job Sat.	Loneliness
<i>Panel A: Full sample</i>				
ATT	0.023 (0.062)	0.028 (0.186)	0.024 (0.027)	0.016* (0.010)
<i>Panel B: Men</i>				
ATT	-0.013 (0.095)			
<i>Panel C: Women</i>				
ATT	0.051 (0.086)			
Pre-trend p	0.878	0.843	—	—
Pre-treatment mean	1.67	48.1	5.38	0.369

Notes: Aggregate ATT estimates from the doubly-robust Callaway–Sant’Anna (2021) estimator, excluding 2020–2021. For GHQ-12 caseness, a higher value indicates worse mental health with the inverse for the SF-12 MCS score. Controls include marital status, having children, home ownership, and education, all measured in 2019. Standard errors in parentheses are bootstrapped (1,000 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 Extension 1: Geographic Heterogeneity

Table 5 and Figure 3 present regional heterogeneity estimates for GHQ-12 caseness. Point estimates vary substantially across regions, though all confidence intervals are wide and cross zero.

The Midlands exhibits the most negative point estimate (-0.274) followed by the devolved nations (-0.057), while the South excluding London (0.217) and the North (0.114) show positive estimates. London’s estimate is almost negligible (0.033), consistent with the hypothesis that metropolitan workers can more easily maintain social networks outside the office. The range of point estimates spans approximately 0.5 GHQ-12 caseness points, suggesting meaningful geographic heterogeneity that is obscured in the aggregate. We interpret these findings as indicative rather than conclusive given the limited precision at the regional level.

Table 5: Geographic Heterogeneity: Effect on Mental Health (GHQ-12) by Region

Region	ATT	SE	95% CI	N
South (excl. London)	0.217	0.113	[−0.004, 0.438]	27,880
North	0.114	0.126	[−0.134, 0.362]	20,630
London	0.033	0.198	[−0.355, 0.421]	10,429
Devolved nations	−0.057	0.145	[−0.341, 0.227]	19,809
Midlands	−0.274	0.183	[−0.633, 0.085]	14,780

Notes: Each row reports the aggregate ATT from a separate doubly-robust Callaway–Sant’Anna (2021) estimator on the regional subsample. North = North East, North West, Yorkshire and the Humber; Midlands = East and West Midlands; South (excl. London) = East of England, South East, South West; Devolved = Wales, Scotland, Northern Ireland. Controls as in Table 4. All estimates exclude 2020–2021. Standard errors bootstrapped (1,000 replications). 95% CIs: $ATT \pm 1.96 \times SE$.

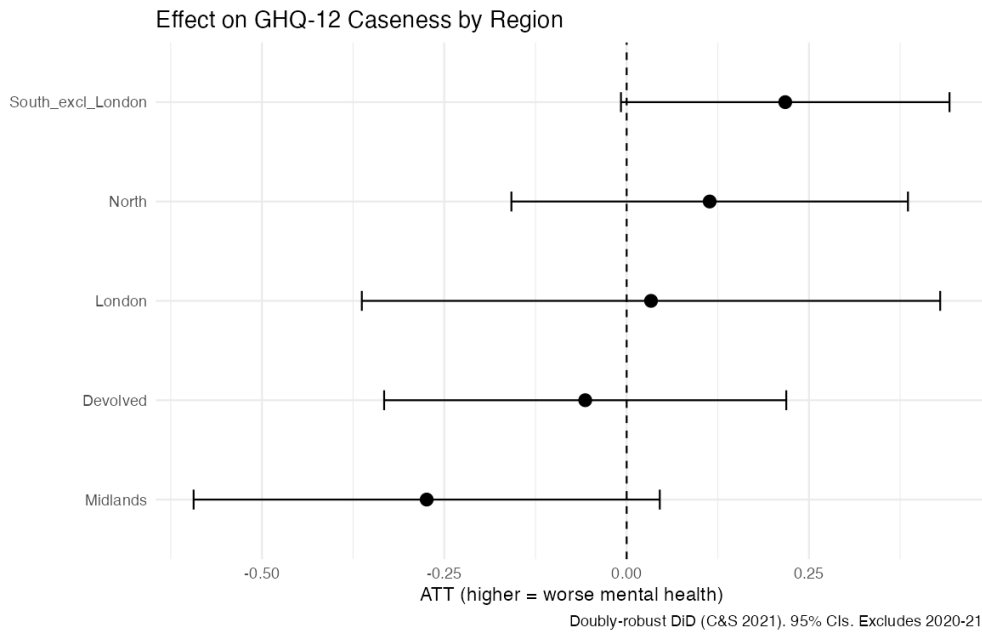


Figure 3: Geographic heterogeneity: ATT estimates for GHQ-12 caseness by region.

Notes: Points are aggregate ATT estimates; horizontal lines are 95% bootstrapped confidence intervals. Regions ordered by point estimate. All estimates exclude 2020–2021.

5.3 Extension 2: Within-Gender Heterogeneity

Table 6 and Figure 4 present within-gender heterogeneity estimates. Disaggregating by partnership and parental status reveals substantial variation in point estimates across subgroups, though confidence intervals are wide given the smaller subgroup sample sizes.

Among women, single mothers exhibit the largest positive point estimate in the GHQ-12 (0.376), consistent with the dual burden hypothesis: single mothers face heightened domestic responsibilities without a partner to buffer the disruptions associated with working from home. Partnered mothers (−0.049) and partnered women without children (0.032) show estimates close to zero. What is notable is that among men, partnered fathers show a positive point estimate (0.174), while partnered men without children (−0.103) and single men without children (−0.164) show negative estimates, suggesting remote work may have been mildly beneficial for these latter groups. Across all subgroups confidence intervals overlap substantially, and we interpret these results as suggestive

of the mechanisms through which social embeddedness conditions the mental health consequences of remote work.

Table 6: Within-Gender Heterogeneity: Effect on Mental Health and Job Satisfaction

Group	GHQ-12		SF-12 MCS		Job Sat.	
	ATT	SE	ATT	SE	ATT	SE
<i>Women</i>						
Single mother	0.376	(0.433)	−0.242	(1.364)	−0.117	(0.195)
Partnered mother	−0.049	(0.157)	0.201	(0.474)	0.006	(0.066)
Partnered, no children	0.032	(0.141)	−0.182	(0.373)	−0.003	(0.057)
Single, no children	−0.065	(0.186)	0.529	(0.532)	0.017	(0.089)
<i>Men</i>						
Partnered father	0.174	(0.156)	−0.335	(0.457)	0.134*	(0.070)
Partnered, no children	−0.103	(0.138)	0.193	(0.422)	−0.051	(0.066)
Single, no children	−0.164	(0.221)	1.277*	(0.725)	0.045	(0.099)

Notes: Each row reports aggregate ATT estimates from separate doubly-robust Callaway–Sant’Anna (2021) estimators for each outcome, on the subgroup defined by gender, partnership status, and parental status as of 2019. GHQ-12: higher = worse mental health. SF-12 MCS: higher = better mental health. Job satisfaction: 1–7. Single fathers excluded ($n = 452$). Controls include education and home ownership measured in 2019. All estimates exclude 2020–2021. Standard errors bootstrapped (1,000 replications). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

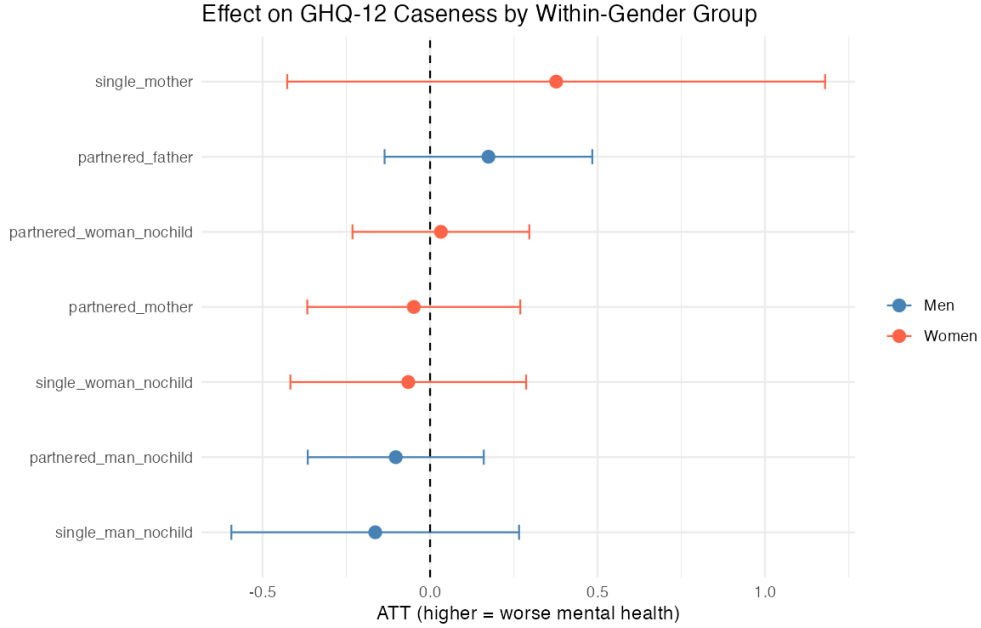


Figure 4: Within-gender heterogeneity: ATT estimates for GHQ-12 caseness by gender-by-partnership-by-parenthood subgroup.

Notes: Points are aggregate ATT estimates; horizontal lines are 95% bootstrapped confidence intervals. Single fathers excluded due to small sample size. All estimates exclude 2020–2021.

5.4 Robustness Checks

We conduct four robustness checks to assess the validity and sensitivity of our main findings, reported in Table 7. The placebo test uses the SF-12 Physical Component Score as an outcome. Since remote work changes where one works rather than physical health directly, a valid identification strategy should produce a null effect on this measure. We find a positive and statistically significant ATT of 0.377 (SE = 0.144). However, a pre-trend p-value of 0.855 confirms that the two groups

followed parallel trajectories in physical health prior to 2020, ruling out a parallel trends violation. We interpret the post-2020 divergence as reflecting greater Covid-related physical health risks faced by non-teleworkable workers who remained in physical workplaces throughout the pandemic, rather than any failure of identification. RC2 restricts the sample to individuals observed in all ten waves. The balanced panel estimate of 0.028 is virtually identical to the main result of 0.023, suggesting differential attrition does not bias our findings. RC3 replaces the continuous GHQ-12 caseness score with a binary indicator for high caseness (GHQ-12 ≥ 8), finding a negligible ATT of 0.005, consistent with effects being absent across the full distribution rather than concentrated among severely distressed workers. RC4 employs a stricter loneliness threshold using only the “often lonely” indicator, yielding an ATT of 0.001, suggesting the marginally significant loneliness effect in the main analysis reflects mild rather than severe isolation.

Table 7: Robustness Checks: Alternative Outcome Specifications and Sample Restrictions

Check	Outcome	ATT	SE	95% CI
Main result	GHQ-12 caseness	0.023	0.062	[−0.099, 0.145]
RC1: Placebo	SF-12 physical score	0.377**	0.144	[0.094, 0.659]
RC2: Balanced panel	GHQ-12 caseness	0.028	0.081	[−0.131, 0.186]
RC3: High caseness	GHQ-12 ≥ 8	0.005	0.006	[−0.006, 0.017]
RC4: Often lonely	Lonely (often only)	0.001	0.005	[−0.009, 0.011]

Notes: All estimates from the doubly-robust Callaway–Sant’Anna (2021) estimator, excluding 2020–2021. The RC1 pre-trend p-value of 0.855 confirms parallel trends hold for the placebo outcome; the significant post-2020 physical health effect reflects greater Covid-related workplace exposure among non-teleworkable workers. 95% CIs: $ATT \pm 1.96 \times SE$. Standard errors bootstrapped (1,000 replications). ** $p < 0.05$.

6 Policy Implications and Limitations

6.1 Policy Implications

The findings of this paper carry implications that extend beyond the academic literature on remote work and wellbeing. If heterogeneous effects by region and within-gender subgroup are borne out, policymakers and employers cannot treat the post-pandemic shift to flexible working as a uniformly benign development whose distributional consequences require no further attention. The aggregate null results documented by Cowan and Spearing (2026) may be precisely that: an aggregate, obscuring concentrated harms among workers whose social circumstances leave them most exposed to the isolating dimensions of home-based work.

For regional policy, workers in the South and North exhibit larger positive point estimates for GHQ-12 caseness than their London counterparts, consistent with metropolitan workers more easily maintaining social networks outside the office. Where this reflects a causal mechanism rather than sampling variation, flexible working mandates designed without geographic targeting risk compounding existing regional inequalities in wellbeing. Employers might consider investing more heavily in structured in-person contact and co-working provision in areas where workplace social networks are less easily substituted, whilst governments should be cautious about treating flexible working expansion as straightforwardly redistributive.

Within-gender heterogeneity findings carry analogous implications for employer-level policy design. The literature has long noted that flexible working affects men and women differently depending on

household composition (Arntz et al., 2022; Agnoletto and Bonomi, 2024), but the policy response has tended towards one-size-fits-all entitlements that do not account for the distinct pressures facing different household configurations. Our results point to single mothers as the subgroup most exposed to mental health deterioration under remote work, consistent with the dual burden of heightened domestic responsibility in the absence of a partner, whilst single childless workers show negative point estimates for GHQ-12 caseness, suggesting remote work may have been mildly beneficial for those without competing household demands. These patterns argue for targeted mental health support embedded within remote work packages rather than offered as a separate opt-in, and for return-to-office provisions attentive to which workers most benefit from in-person contact.

More broadly, this paper contributes to a growing recognition that the design of post-pandemic labour market institutions requires a more granular evidence base than aggregate average effects can provide. Understanding which workers, in which places, and under which household circumstances bear the largest mental health costs of remote work is a precondition for policy that is both effective and equitable. The framework developed here offers one route toward that understanding.

6.2 Limitations

Our analysis faces several data and methodological constraints. Firstly, the post-treatment observation window is compressed: after excluding the acute pandemic years 2020–2021, only two post-treatment years (2022–2023) remain, limiting our ability to detect more persistent consequences of the work-from-home revolution.

Additionally, all wellbeing measures rely on self-reported outcomes, which are subject to recall bias and potential changes in reporting behaviour during the pandemic. However, GHQ-12 and SF-12 are well-validated instruments widely used in the economics literature, and the consistency of our null mental health findings across both measures reduces the likelihood that reporting artefacts are driving results. Further, the loneliness outcome is available only from wave 9 (2017–18) onwards, providing a shorter pre-treatment window; however, the available pre-treatment years show no evidence of differential trends, supporting identification for this outcome.

Household composition is fixed at 2019 values throughout. This is a deliberate design choice that prevents the conditioning variables from being contaminated by the treatment, though it means covariates may imperfectly reflect actual household circumstances in later waves. And finally (on data limitations), the sample is restricted to employed workers aged 25–64, which excludes potentially vulnerable groups; however, this restriction is standard in the remote work literature and ensures treatment and control groups face comparable labour market conditions.

Methodologically, the heterogeneity analysis amplifies statistical power constraints: point estimates diverge substantially across regions and gender subgroups, yet confidence intervals are uniformly wide, rendering most subgroup estimates individually insignificant. We therefore interpret the heterogeneity results as indicative of the mechanisms through which social embeddedness conditions the wellbeing consequences of remote work, rather than as precise causal estimates. Single fathers (n=452) are excluded from the within-gender analysis entirely due to insufficient sample size.

Finally, the conditional parallel trends assumption may be violated if occupational composition

within teleworkability categories shifted differentially across regions or household types in ways correlated with pre-existing wellbeing trends; however, the balance tests and pre-trend evidence in Section 4.3 provide reassurance that the assumption holds in aggregate.

Several avenues for future research emerge from these limitations. As longer post-pandemic panels become available, future work could examine whether the null mental health effects documented here persist over a longer horizon or whether more gradual consequences of sustained remote work emerge over time. Individual-level data on workplace social networks and peer interactions would allow researchers to directly test the social embeddedness mechanism proposed here. Larger subgroup samples would enable more precise inference on geographic and within-gender heterogeneity, potentially transforming the suggestive patterns we document into statistically reliable estimates. Finally, complementing self-reported wellbeing measures with objective health outcomes, such as biomarker data or administrative health records, would address concerns about reporting bias and provide a fuller picture of the consequences of the work-from-home revolution (Crossley and Kennedy, 2002; Ferrer-i Carbonell and Frijters, 2004).

7 Conclusion

This paper set out to examine whether the aggregate null effects of remote work on wellbeing documented by Cowan and Spearing (2026) conceal meaningful heterogeneity across UK regions and within gender subgroups defined by partnership and parental status. Applying the doubly robust differences-in-differences estimator of Callaway and Sant’Anna (2021) to UKHLS data spanning 2014 to 2023, and exploiting pre-pandemic occupational teleworkability as a source of plausibly exogenous variation, we find suggestive evidence that they do. Single mothers exhibit the largest adverse point estimates for GHQ-12 caseness, again consistent with the dual burden of heightened domestic responsibility, with single childless workers of either gender showing negative estimates. Geographically, workers in the South and North show worse mental health trajectories than those in London, consistent with the hypothesis that metropolitan labour markets offer denser social networks that partially substitute for reduced workplace contact. Wide confidence intervals across all subgroup estimates counsel against strong conclusions, and so we interpret these findings as indicative rather than definitive.

The paper makes two contributions to a literature that has largely stopped at aggregate average effects. We provide the first systematic evidence on geographic heterogeneity in the remote work–wellbeing relationship within the UK, and we extend within-gender analysis to account for the interaction of partnership and parental status, moving beyond the coarse male/female comparison that dominates existing work. Taken together, the results underscore that the welfare consequences of the post-pandemic shift to flexible working are not uniformly distributed: they depend on who is working remotely, where they live, and what social infrastructure surrounds them outside the office. Whether these patterns are robust to alternative specifications and larger subgroup samples remains an important question for future work, particularly as longer post-pandemic panel data become available.

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